OPERATIONAL EXCELLENCE · WHAT THE RESEARCH SAYS

The fuller a hospital runs, the less room it has for the next emergency.

The trouble starts well before it is full.

A British simulation and data on 2.65 million Danish admissions show why spare capacity belongs in the design.

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THE PROBLEM

Why a reasonable average can still hide danger

Emergencies don't arrive evenly. Some days are quiet, and others bring a rush. A hospital that runs close to capacity has no spare beds to absorb the rush, so one busy day can turn into a crisis.

Bed occupancy is the share of available beds that are in use.

STUDY 1

What the British model found

Researchers built a computer simulation of a hospital that takes emergency admissions. The BMJ published the results in 1999.

85%

Below this average, risk stayed low. At this level, the model expected the hospital to run short of beds on about four days a year.

90%

At this level or higher, bed crises became a regular event.

THE HIDDEN COST

Four days a year sounds small. The damage lasts much longer.

8 weeks

The study estimated that a hospital that runs out of beds on four days a year may see its work disrupted for up to eight weeks in total.

STUDY 2

9%

higher mortality linked to high bed occupancy

Danish researchers analyzed 2.65 million admissions to medical departments from 1995 to 2012. High occupancy was linked to a 9% rise in in-hospital and 30-day mortality, compared with low occupancy.

This is a statistical association. On its own, it does not prove that occupancy was the cause.

BEYOND HOSPITALS

The same pattern shows up far from the hospital ward

Any system with uneven demand behaves in a similar way, whether it is a call center, a production line, or a team that handles requests. As it nears full capacity, queues grow much faster than the workload.

This point comes from queueing theory, the math of waiting lines. It is not a finding of the two studies.

WHAT I TAKE FROM THIS FOR LEADERS

Spare capacity is part of the design, not waste.

1

Track occupancy at peak times, not only the yearly average. The average hides the days when the system breaks.

2

Plan capacity for the swings in demand, not just the average.

3

Don't judge an initiative on one year at one site. The British study found that random variation swamps the effect.

> If your team is busy all the time, where will tomorrow's urgent work go?

SOURCES AND LIMITS

What this evidence can and cannot tell us

The British study is a simulation of a hypothetical hospital, not real hospital data, and it dates from 1999.

The Danish study shows an association, not cause, and covers medical departments in one country.

The 85% figure comes from one model. The right limit can differ from one hospital to another.

Editorial disclaimer: The views and analyses presented in this article represent the author's professional interpretation based on the cited sources and do not necessarily represent the official positions of the organizations mentioned. For the original, complete and most current information, please consult the primary sources and official documents listed in the references.

[1] Bagust, A., Place, M., & Posnett, J. W. (1999). Dynamics of bed use in accommodating emergency admissions: stochastic simulation model. *BMJ*, 319(7203), 155–158.

[2] Madsen, F., Ladelund, S., & Linneberg, A. (2014). High levels of bed occupancy associated with increased inpatient and thirty-day hospital mortality in Denmark. *Health Affairs*, 33(7), 1236–1244.

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